

CanConstructNZ Smart System

International Workshop; Validation and Academic–Industry Engagement Report

34th Annual Conference of the International Group for Lean Construction (IGLC 34)
43rd International Symposium on Automation and Robotics in Construction (ISARC 2026)

National University of Singapore (NUS), Singapore | 22 June 2026

Presenters: Associate Professor Mostafa Babaeian Jelodar, Team Lead - Smart System | Professor Monty Sutrisna, Primary Investigator and Scientific Lead

Report purpose

To document the workshop as evidence of international stakeholder engagement and preliminary validation of the CanConstructNZ smart system, with particular emphasis on whether the system captures the construction-sector problems that matter to practitioners and whether its outputs are viewed as useful for planning and decision-making.

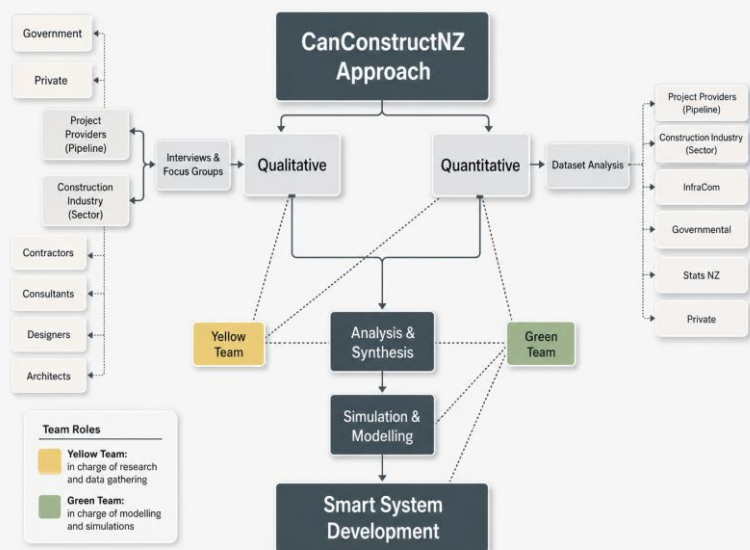
Evidence framing: stakeholder / face validation and user-needs validation

Brief introduction

CanConstructNZ brings together qualitative stakeholder evidence and quantitative construction-sector data, then connects analysis and synthesis with simulation and modelling to support development of the Smart System.

The workshop positioned this integrated approach around pipeline and sector forecasting, shocks and stresses, agent-based modelling and dashboard-based decision support. The core proposition is that better visibility of future demand, sector constraints and contractor capacity can support more reliable portfolio planning and sequencing.

Workshop focus: testing whether the modelling logic and emerging Smart System reflect the uncertainties, constraints and decisions recognised by academic and industry participants.



CanConstructNZ approach linking qualitative and quantitative evidence to analysis, simulation and Smart System development (workshop presentation).

Executive Summary

The CanConstructNZ workshop held on 22 June 2026 at the 34th Annual Conference of the International Group for Lean Construction (IGLC 34) and the 43rd International Symposium on Automation and Robotics in Construction (ISARC 2026), at the National University of Singapore (NUS), provided a high-value international forum for presenting, testing and discussing the CanConstructNZ modelling approach and smart system. The workshop combined a detailed explanation of the construction-pipeline and sector-capacity models with interactive participant questions focused on uncertainty, resource constraints, contractor behaviour and the usefulness of the smart system.

The evidence supports a strong case that the workshop acted as a meaningful stakeholder-validation exercise. Participants identified project timing and funding uncertainty as the dominant form of pipeline uncertainty (71%), followed by external shocks and market disruptions (47%). When asked what first constrains the sector as pipeline activity increases, the strongest recurring concepts were resources, labour, time and cost. These responses directly align with the system's core design logic: forecasting future demand, modelling shocks, generating project workloads, reconstructing contractor populations, matching projects to available contractor capacity, and showing capacity or deficit through an interactive dashboard.

71%	4.2 / 5	3.9 / 5	4.0 / 5
Project timing / funding uncertainty	Usefulness for understanding capacity & capability	Capturing regional & market constraints	Usefulness for guiding further development

Participants also rated the smart system positively as a decision-support proposition. Its highest usefulness rating was for understanding sector capacity and capability (4.2/5). The system was also rated positively for guiding future development (4.0/5), supporting planning and decision-making (3.7/5), and testing pipeline uncertainty scenarios (3.5/5). Effectiveness ratings were strongest for regional and market constraints (3.9/5), followed by project-pipeline uncertainty (3.7/5). Main-contractor capacity received a more moderate rating (3.2/5), providing a clear development signal rather than a negative result.

Overall validation statement

The workshop provides credible international stakeholder evidence that the CanConstructNZ smart system is addressing the right sector problem, representing several of the right system variables, and offering a useful basis for capacity-and-capability planning. It also identified a specific next-stage requirement: strengthen the behavioural representation of contractor project-selection decisions, particularly risk, profitability, client reliability and internal capacity.

This evidence should be presented as stakeholder, face and relevance validation. It complements—but does not replace—technical model verification, statistical forecasting validation, sensitivity testing or out-of-sample performance assessment.

Workshop Context and Engagement Design

The workshop, titled “Modelling for Construction Capacity and Capability: Portfolio Planning, Lean Philosophy, and Lessons from CanConstructNZ”, was delivered on 22 June 2026 at the 34th Annual Conference of the International Group for Lean Construction (IGLC 34) and the 43rd International Symposium on Automation and Robotics in Construction (ISARC 2026), hosted at the National University of Singapore (NUS). It was presented by lead speaker Associate Professor Mostafa Babaeian Jelodar, Team Lead - Smart System, and Professor Monty Sutrisna, Primary Investigator and Scientific Lead. Its structure deliberately connected the CanConstructNZ research programme to both lean-construction and automation/modelling audiences. The presentation covered the programme context, construction-sector complexities, pipeline and sector modelling, the smart system, key takeaways and the Modern Methods of Construction and Automation in Built Environment (MOCABE) research environment.

A deliberately mixed academic and industry audience

The interactive engagement results show a heterogeneous audience rather than a single stakeholder group. Participants described roles or organisational perspectives including project management, education, lean consulting and leadership, management, builders, developers, civil design, software, building information modelling (BIM), productivity, supply-chain monitoring, smart manufacturing, research, safety and compliance. This breadth is important because the system is intended to operate at the boundary between data modelling, portfolio planning and sector decision-making.

What best describes your organisation's and your role in the construction industry?



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Figure 1. Participant roles and organisational perspectives captured during the workshop (Mentimeter).

Why this matters for validation

A smart system for construction-sector capacity and capability needs to be understandable and relevant across client, contractor, consultant, academic and technology perspectives. The participant mix therefore strengthens the workshop's value as a cross-sector relevance check rather than a presentation to a narrowly defined specialist group.

The CanConstructNZ Smart System Presented to Participants

The workshop presented CanConstructNZ as an integrated forecasting, simulation and decision-support system. The architecture links pipeline projections and sector projections to agent-based allocation logic. Projects are ranked and represented as pipeline agents; sector firms are represented as contractor agents with capacity and workload characteristics; projects are allocated to main contractors and subcontractors; agents are updated as allocation occurs; and the system calculates both project allocation outcomes and sector capacity/capability outcomes for dashboard presentation.

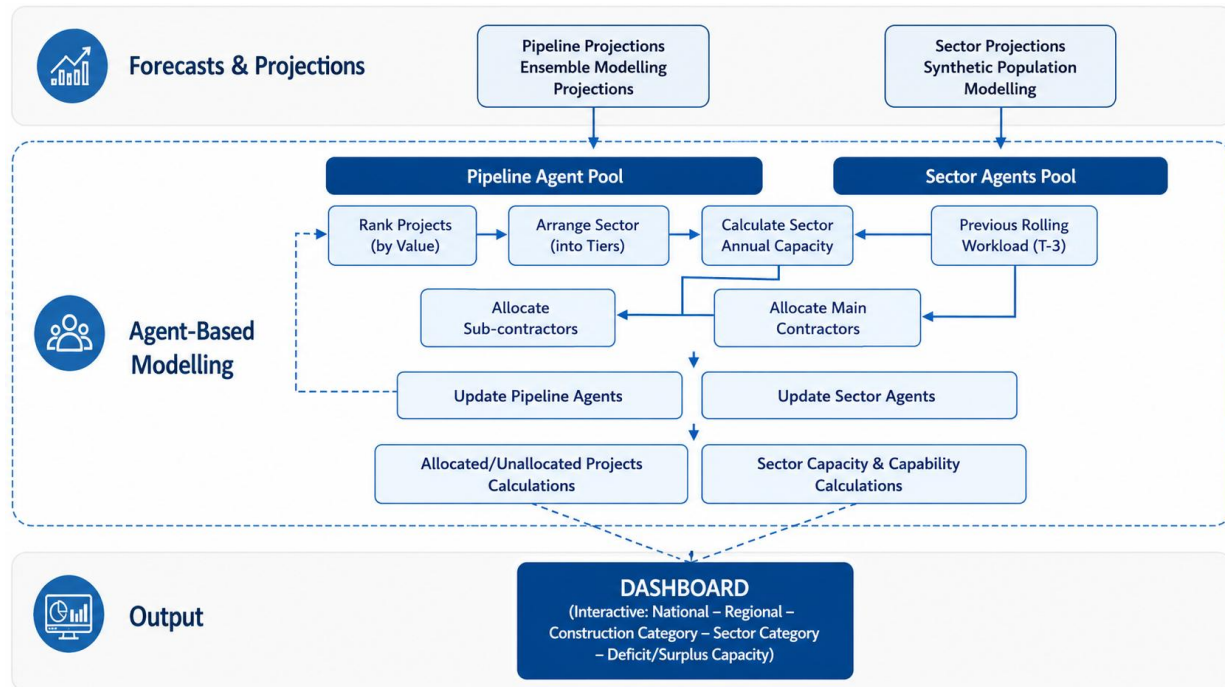


Figure 2. Smart-system architecture presented at the workshop: forecasts and projections feed agent pools, allocation logic and a dashboard output.

Core modelling layers communicated in the workshop

Layer	Decision question	Approach presented
Pipeline forecast	How many projects may emerge, and what is their total value?	Ensemble modelling using GAM, ECM and robust Huber regression.
Shock overlay	How do shocks and policy changes alter the forecast?	Composite-index and time-series / system-dynamics approaches.
Project instances & duration	What projects are likely, where, at what value, and for how long?	Probabilistic sampling / Monte Carlo and probability-density modelling.
Workload & contractor capacity	What workload is created and what delivery capacity exists?	Agent-based modelling and synthetic population reconstruction.
Project-contractor matching	Which contractors can realistically deliver which projects?	Rule-based agent-based matching using project and contractor constraints.

From Forecasting to Portfolio Intelligence

A distinctive feature of the workshop was the translation of modelling outputs into portfolio-level planning logic. The presentation framed construction pipelines as flows of work through a constrained delivery system. Poor visibility of future demand creates waste through delay, overloading, under-utilisation and reactive decision-making; accordingly, capacity and capability should be understood before large portfolios of work are committed.

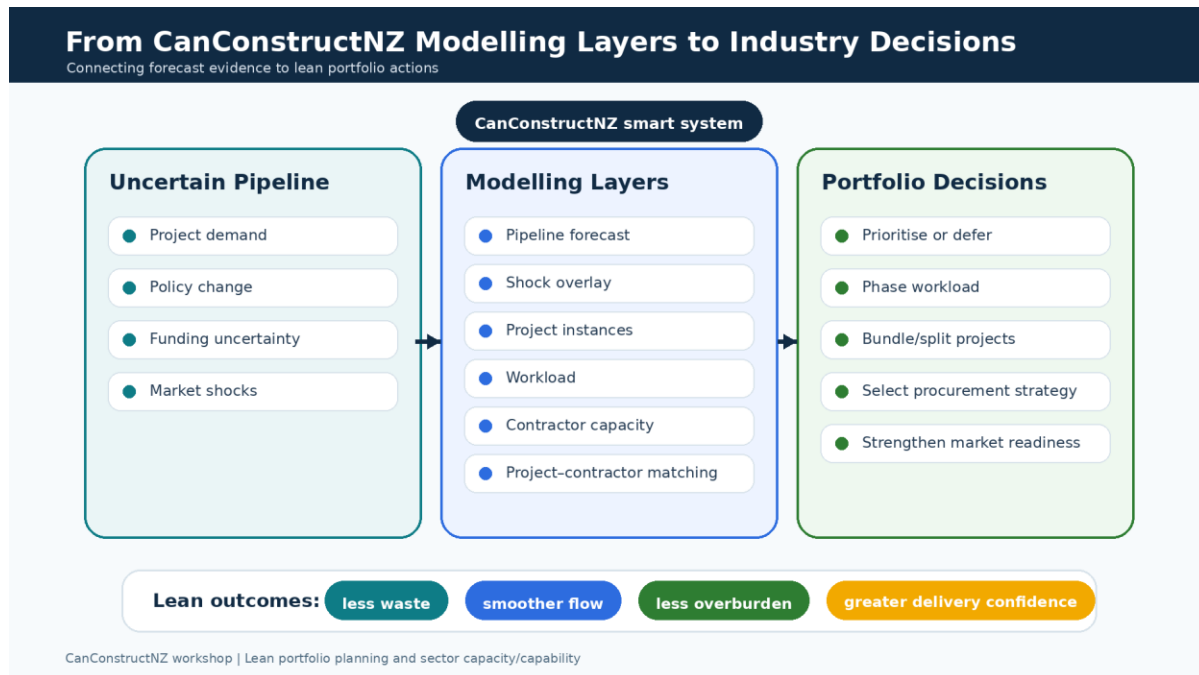


Figure 3. Workshop framing of CanConstructNZ modelling layers as a bridge from uncertain pipeline conditions to portfolio decisions and lean outcomes.

The decision loop tested with participants

The workshop's key-takeaway sequence described a recurring decision loop: forecast the future pipeline; simulate project instances and durations; match projects to contractor capacity; deliver and sequence the portfolio; learn from observed outcomes; and improve rules, data and decisions. This is important for validation because it moves the system away from being a static forecasting dashboard and toward a learning-oriented portfolio-intelligence platform.

- Forecasting creates visibility of the future pipeline rather than relying only on current committed work.
- Simulation makes uncertainty explicit by representing project timing, value, duration and shock scenarios.
- Matching connects demand to realistic sector capacity and capability rather than treating the sector as unconstrained.
- Portfolio outputs can support prioritisation, phasing, bundling/splitting and procurement strategy decisions.
- Learning from outcomes creates a pathway for continuous improvement of data, behavioural rules and system calibration.

Validation proposition tested in the room

If practitioners identify uncertainty, resource constraints and contractor decision behaviour as the main drivers of delivery reliability—and the system explicitly models those issues—then the workshop provides a direct test of problem–solution alignment.

Stakeholder Evidence: Validation of the Industry Problem

Participant responses strongly reinforced the problem definition underpinning CanConstructNZ. The most important uncertainty was not simply the total amount of demand, but whether projects actually start when expected and whether funding and procurement remain reliable. Seventy-one percent selected project timing and funding uncertainty, compared with 47% for external shocks and market disruptions and 24% for demand fluctuation. The multiple-response pattern also indicates that uncertainty is layered: pipeline timing, market shocks and demand changes coexist rather than operating as separate issues.

Mentimeter

What type of pipeline uncertainty has the biggest impact on your organisation?

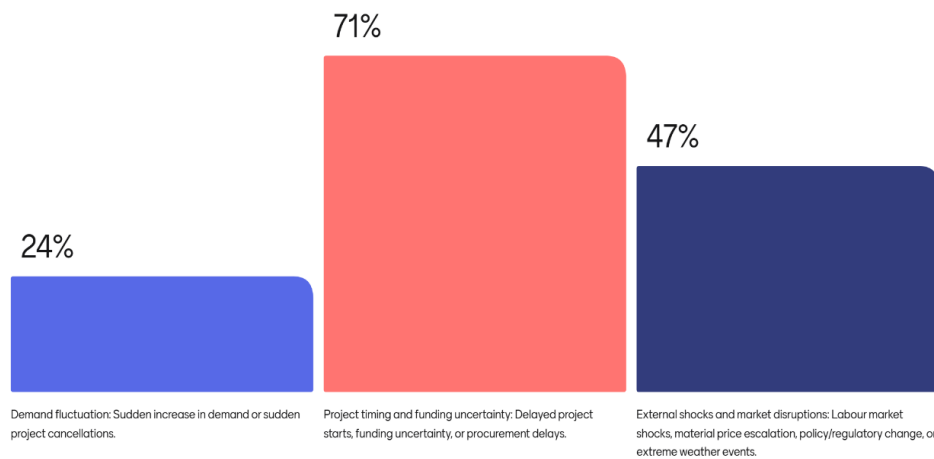


Figure 4. Participant views on the forms of pipeline uncertainty with the greatest organisational impact.

Contractor behaviour confirms the need for more than a capacity-only allocation rule

When participants ranked the factors that most affect whether a main contractor takes on a project, risk profile ranked first, followed by margin/profitability, client reliability and available internal capacity. Supply-chain availability, contract type, project size and location followed. This is a particularly useful validation result: it supports the model's inclusion of capacity, size, specialisation and location, while also identifying behavioural and commercial variables that should be incorporated more explicitly in future versions of the project-contractor matching logic.

Rank	Factor	Implication for CanConstructNZ
1	Risk profile	Introduce or refine contractor risk-tolerance and project-risk attributes.
2	Margin / profitability	Test commercial attractiveness as a project-acceptance rule.
3	Client reliability	Consider client / counterparty quality as a behavioural decision variable.
4	Available internal capacity	Directly validates the central capacity constraint in the model.
5–8	Supply chain, contract type, project size, location	Use as secondary matching or scenario variables where data are available.

Stakeholder Evidence: Validation of the Smart System

Participants were asked directly how effectively the smart system captures key information and how useful it would be for several planning purposes. The ratings are positive overall, but they also differentiate where the system is strongest and where further refinement is required.

Validation dimension	Rating	Interpretation
Capturing regional and market constraints	3.9 / 5	Strongest effectiveness result; supports the regionalised sector-capacity and market-pressure logic.
Capturing construction-pipeline uncertainty	3.7 / 5	Positive face-validity result for the forecast / shock / scenario components.
Capturing main-contractor capacity	3.2 / 5	Moderate result; indicates the need for richer contractor behaviour and capacity representation.
Usefulness: understanding sector capacity & capability	4.2 / 5	Highest usefulness score and a direct endorsement of the system's primary purpose.
Usefulness: guiding future system development	4.0 / 5	Strong evidence that participants saw value in continued development and refinement.
Usefulness: planning and decision-making	3.7 / 5	Positive support for client, contractor and policymaker decision use.
Usefulness: testing pipeline uncertainty scenarios	3.5 / 5	Useful but with scope to strengthen scenario design, communication and interaction.

 Mentimeter

How useful would the smart system be for the following purposes?

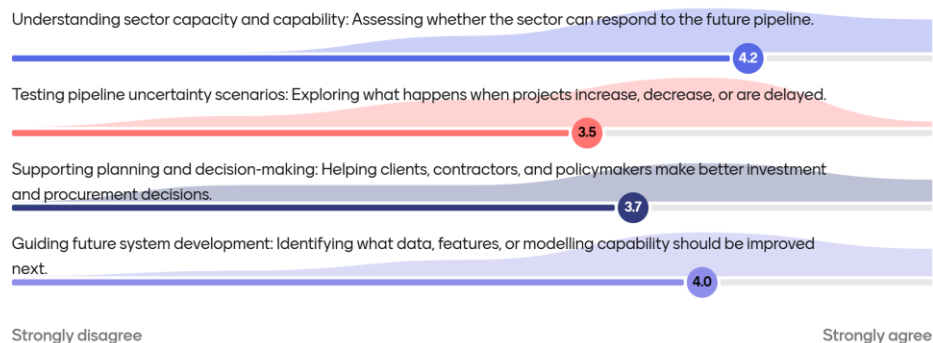


Figure 5. Participant ratings of the potential usefulness of the smart system (1 = strongly disagree; 5 = strongly agree).

Interpretation

The pattern is consistent with a credible prototype or developing decision-support system: participants can see clear value in the purpose and overall architecture, while rating one of the more difficult modelling tasks—contractor capacity and acceptance behaviour—more cautiously. That difference is actionable evidence for development prioritisation.

Validation Matrix and System-Development Priorities

Taken together, the workshop evidence provides both validation and design guidance. The strongest reportable outcome is not that every component is “fully validated”, but that the modelling architecture is well aligned with the sector concerns raised by an international mixed audience, and that participant ratings identify a clear hierarchy for refinement.

Stakeholder signal	Workshop evidence	System component	Validation / action
Pipeline timing / funding uncertainty	71% identified it as the largest uncertainty impact.	Pipeline forecast; shock overlay; scenario inputs; project instances.	Strong problem–model alignment. Continue improving scenario timing and policy/funding representations.
External shocks / market disruption	47% identified this as a major uncertainty source.	Shock overlay using composite indicators and longer-term structural approaches.	Validates explicit shock modelling rather than assuming a smooth pipeline.
Sector resource constraints	Word cloud emphasised resources, labour, time, cost, capacity and capability.	Workload calculation; contractor capacity; sector agent pool; deficit/surplus dashboard.	Strong face-validity signal for capacity/capability as the central decision lens.
Contractor project-selection behaviour	Risk, profitability, client reliability and internal capacity ranked highest.	Current matching uses capacity and project/contractor attributes.	Highest-priority enhancement: enrich project-acceptance rules with behavioural/commercial factors.
Decision usefulness	4.2/5 for capacity/capability; 3.7/5 for planning; 3.5/5 scenarios.	Interactive dashboard and model outputs.	Evidence of practical relevance; improve scenario usability and decision narratives.
System evolution	4.0/5 usefulness for guiding future development.	Forecast–simulate–match–deliver–learn–improve loop.	Supports a co-development pathway with users rather than a one-off model release.

Recommended development priorities arising directly from the workshop

- Strengthen contractor-agent decision rules to represent risk tolerance, profitability expectations, client reliability and potentially contract form and supply-chain conditions.
- Expand scenario design around project timing, funding delays and procurement disruption because these were more salient to participants than demand-volume fluctuation alone.
- Retain regional and market-constraint modelling as a core strength and continue exposing location-based labour and market pressures in the dashboard.
- Use future workshops to compare simulated contractor/project decisions with observed industry behaviour, turning face validation into behavioural calibration and empirical validation.
- Capture participant type, sample size and role category more systematically in future engagement rounds to allow segmented analysis of client, contractor, consultant and academic perspectives.

Suggested project-reporting language

“The June 2026 IGLC–ISARC workshop provided international stakeholder validation of the CanConstructNZ smart-system concept and modelling architecture. Participant feedback confirmed the salience of pipeline uncertainty, sector resource constraints and contractor decision behaviour, while usefulness ratings demonstrated strong perceived value for understanding sector capacity and capability and for guiding further system development.”

Academic–Industry Engagement, Strategic Value and Conclusions

Evidence of international academic and industry engagement

The workshop itself is a significant engagement outcome because it placed the CanConstructNZ smart system in front of two complementary international communities: lean-construction researchers and practitioners concerned with flow, reliability and portfolio performance, and automation/robotics researchers and practitioners concerned with modelling, simulation and digital systems. The content deliberately bridged these communities by linking pipeline visibility and lean portfolio principles with ensemble forecasting, synthetic population modelling, agent-based simulation and dashboard-based decision support.

The participant role cloud reinforces this cross-disciplinary reach. Contributions came from people identifying with safety, project management, education, lean consulting, construction management, builders, developers, design, BIM, software, supply-chain monitoring, productivity, smart manufacturing and research. The engagement therefore functioned as more than dissemination: participants were asked to interrogate the problem definition, identify decision constraints, rank contractor-selection factors and judge the smart system's effectiveness and usefulness.

Strategic value for CanConstructNZ

- Validation: the workshop provides evidence that the system addresses sector concerns recognised by a diverse international audience.
- Co-development: the lower contractor-capacity score and ranked project-acceptance factors translate directly into an evidence-based development agenda.
- Research translation: modelling methods were connected to practical portfolio decisions rather than presented only as technical methods.
- Academic impact: the workshop linked forecasting, agent-based modelling, lean portfolio planning and digital decision support in an international conference setting.
- Industry relevance: participant feedback focused on operational realities—labour/resources, risk, profitability, funding uncertainty, supply chains and internal capacity—providing a practical test of model assumptions.

Conclusion

The 22 June 2026 workshop provides a strong, defensible validation narrative for the CanConstructNZ smart system. The evidence shows that participants recognised the core problems the system is designed to address and rated its primary value proposition—understanding construction-sector capacity and capability—highly. The strongest effectiveness rating for regional and market constraints indicates that the spatial and market-facing aspects of the system are already persuasive to users. At the same time, the more moderate rating for main-contractor capacity usefully identifies where the next generation of the model should focus.

Accordingly, the workshop should be positioned as international stakeholder validation and academic–industry engagement that has both confirmed the direction of the smart system and generated concrete requirements for its next development cycle. Future validation should build on this base through structured user testing, contractor-behaviour calibration, comparison with observed allocation outcomes, scenario-based case studies and repeat engagement with identifiable stakeholder cohorts.

Final assessment

The workshop validates the relevance and direction of CanConstructNZ, demonstrates meaningful international academic–industry engagement, and provides a clear evidence trail from stakeholder concerns to model architecture, user value and future system improvement.

Source basis: CanConstructNZ workshop presentation, 22 June 2026 (including workshop figures and modelling framework); Participant engagement results (Mentimeter), Joint ISARC–IGLC conference workshop, NUS, Singapore.